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Te Whare Wānanga o Otāgo
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Burst-MAC: A MAC Protocol for Handling Burst Traffic in LoRa Network

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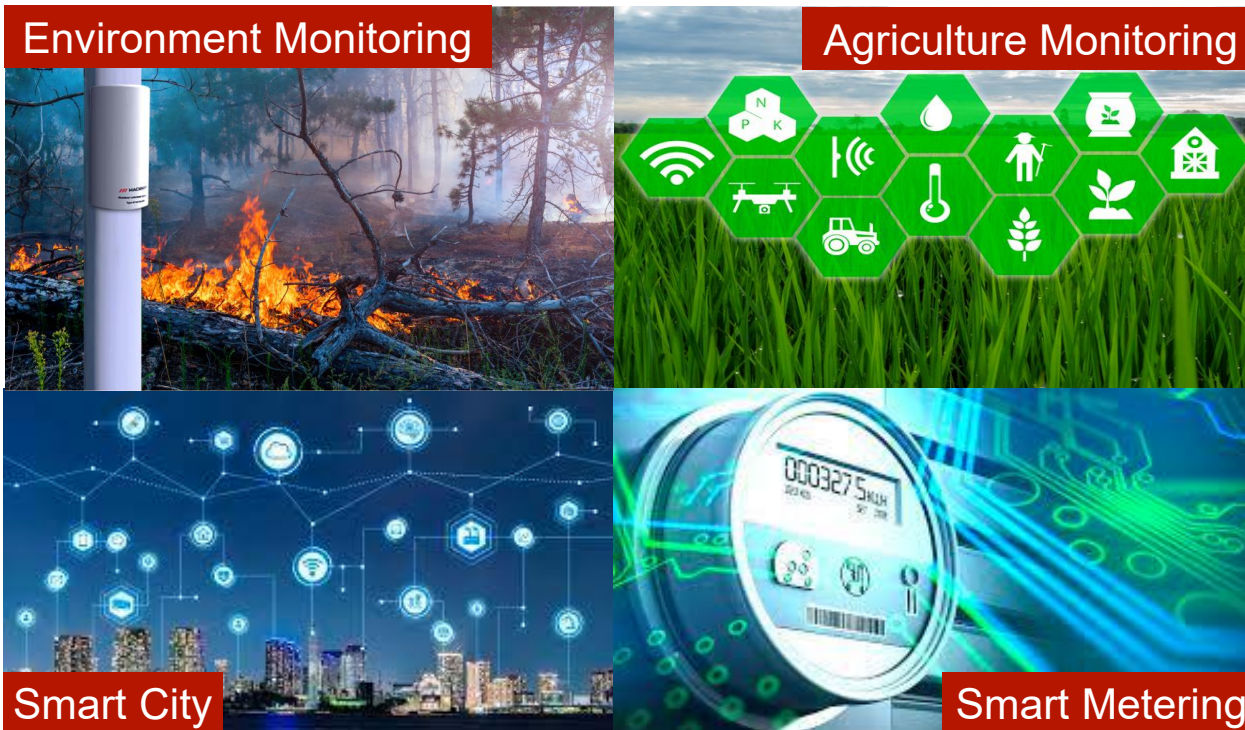
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Low Power Wide Area Network

◆ Low Power Wide Area Network (LPWAN) has enabled various IoT applications.



LoRa[®] (Long Range)

- ◆ Wide coverage
- ◆ High capacity
- ◆ Long battery life
- ◆ Low Cost

5.9 million gateways and **300 million end nodes** with LoRa deployed worldwide, over **600 documented use cases** (May 2024).

LoRa Overview

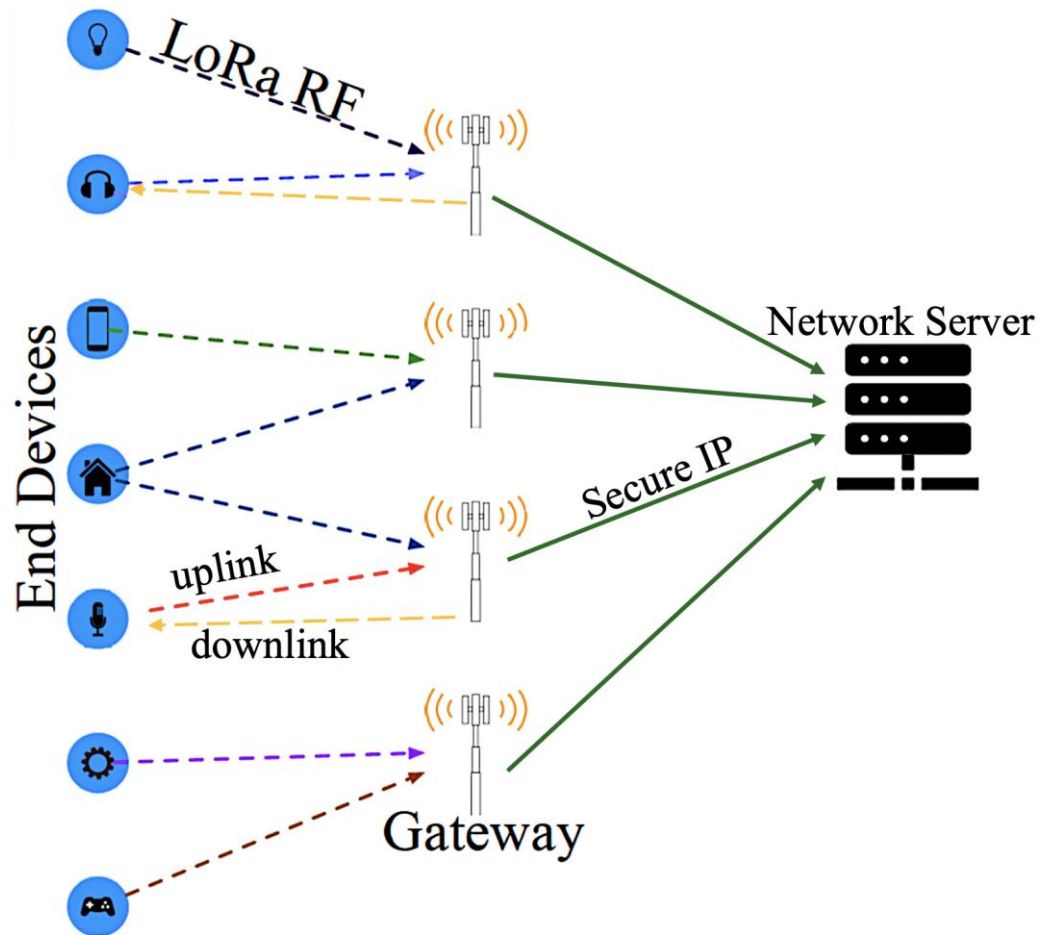


Fig: LoRa network architecture

- ◆ End Devices (nodes), Gateway, Network Server.
- ◆ Communication Path/ Channel:
 - Nodes transmit packets to gateway via *Uplink channel*.
 - Gateway communicates with nodes via *Downlink channel*.
- ◆ Chirp Spread Spectrum (CSS) technology.
- ◆ Spreading Factor (SF): implies the no. of raw bits encoded in a symbol. (SF 7-12)

LoRa Overview

- ◆ LoRa is typically designed to handle very infrequent communications.



Fig: Forest fire monitoring in California forest

What is Burst Traffic?

- ✦ However, many applications experience **Burst Traffic**. E.g., emergencies like **Forrest Fire**, Volcanic Eruption, Earthquake, or Smart Cities during peak hours.

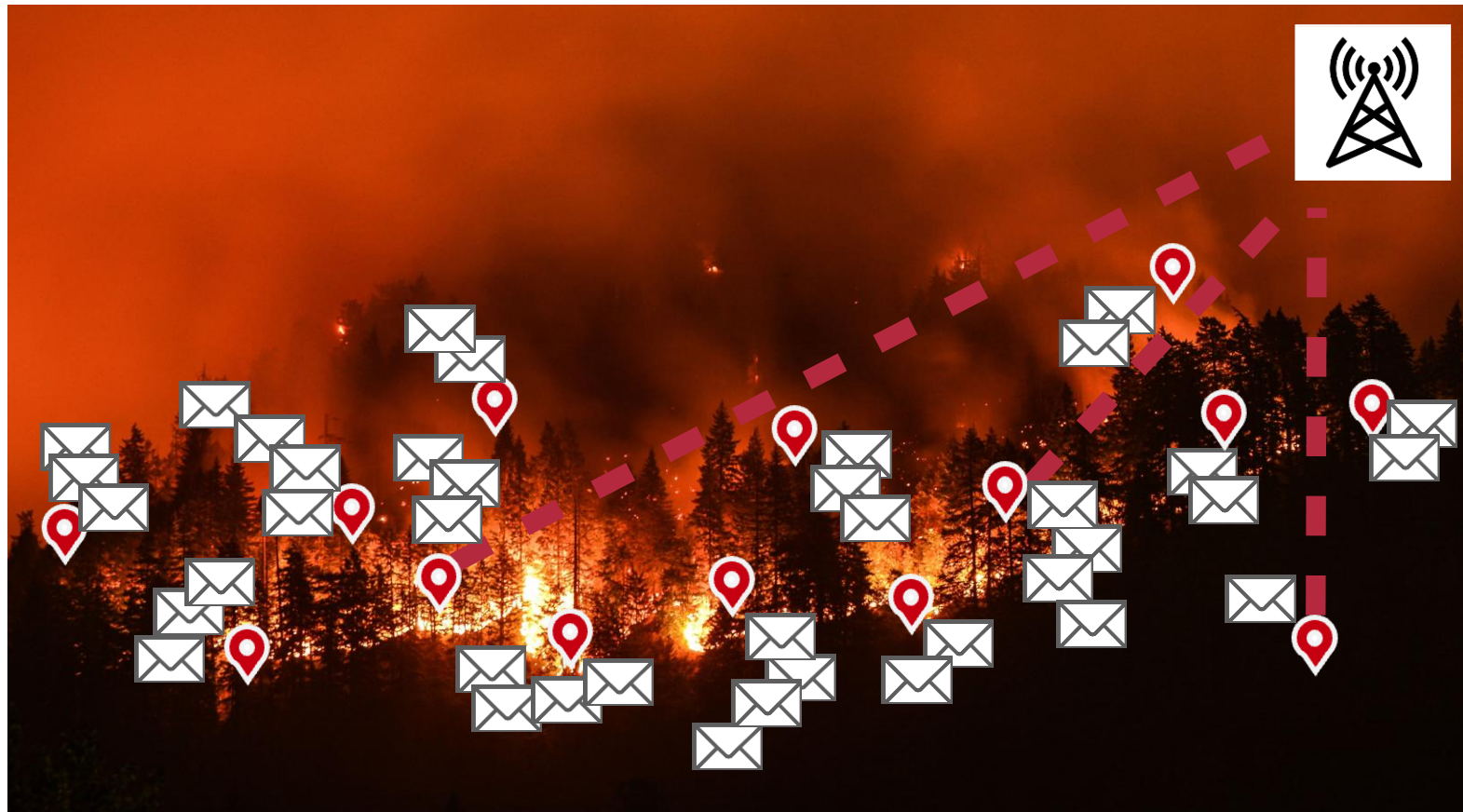
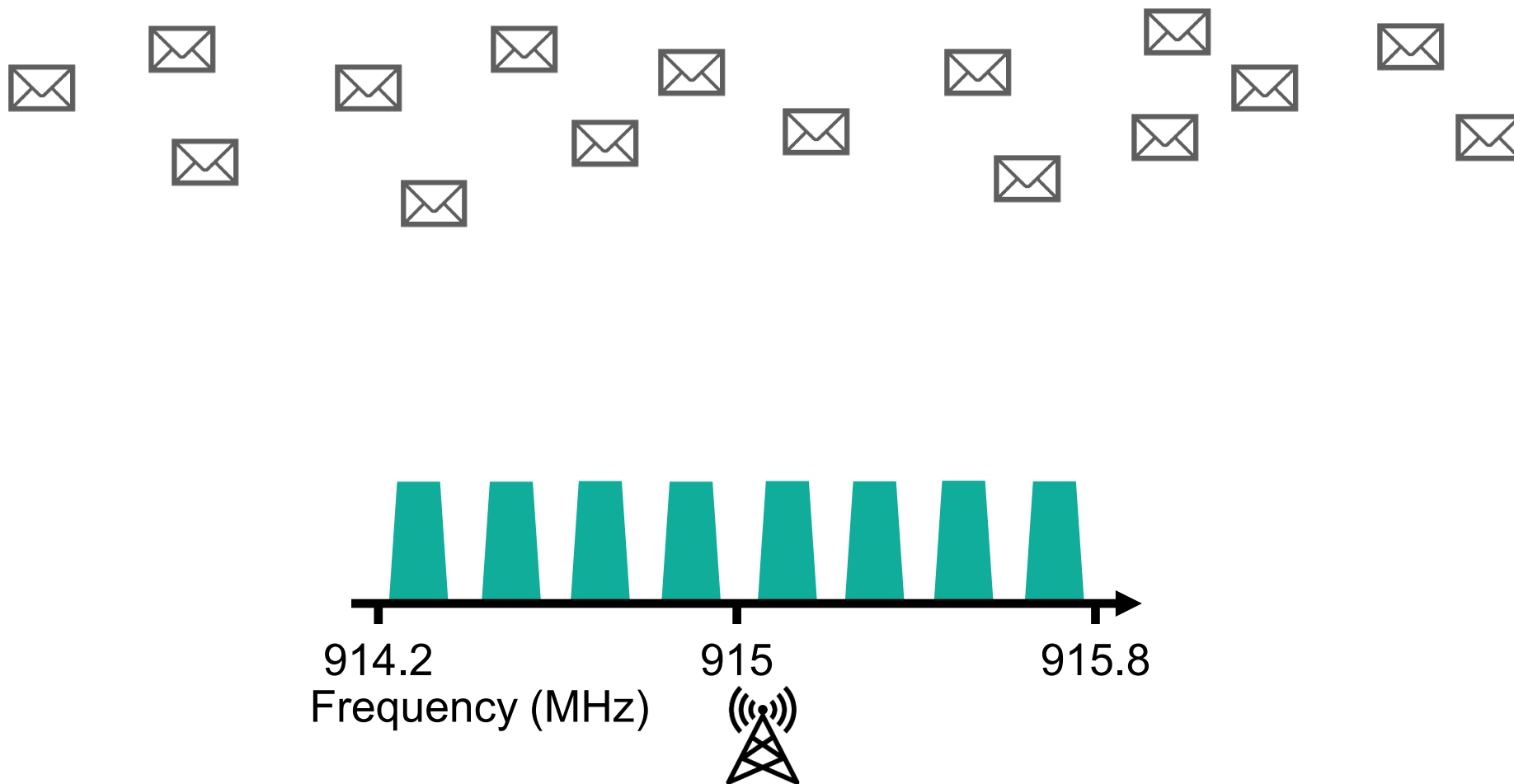


Fig: California Wildfire (2021)

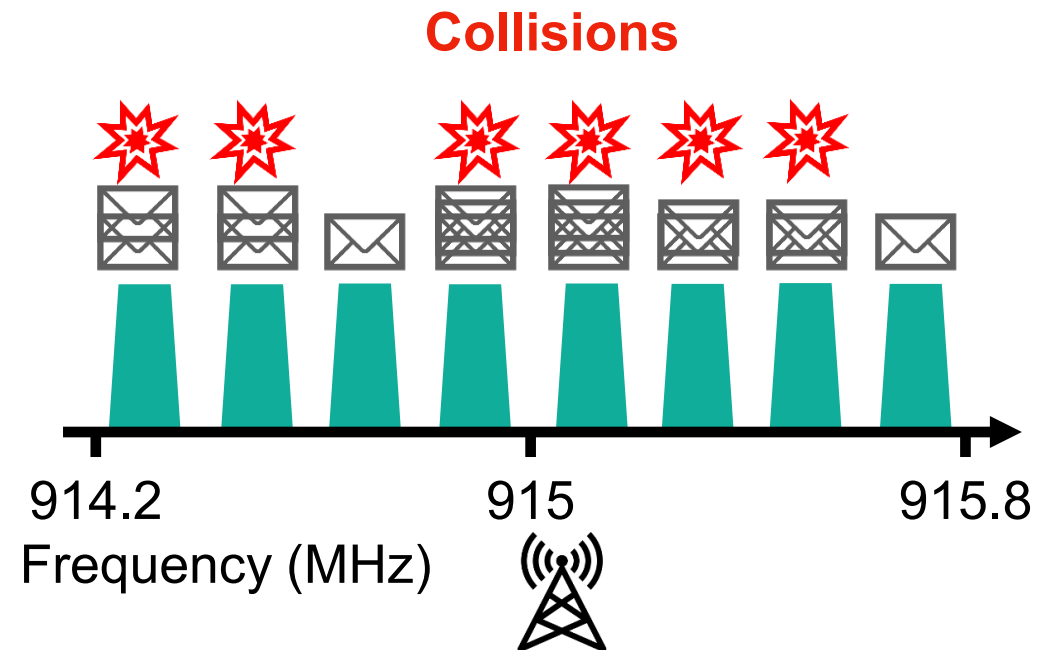
Numerous nodes start transmitting data packets simultaneously in quick succession.

Leads to temporary surge in network activity 

Consequences of Burst Traffic in LoRa



Consequences of Burst Traffic in LoRa



Why is it necessary to handle Burst Traffic?

Current MAC protocol of LoRa:
LoRaWAN
adopts **ALOHA**

NO collision avoidance mechanism!

- Increased packet loss*
- Increased retransmissions
- Rapid battery depletion
- Excessive network latencies
- Degraded throughput*

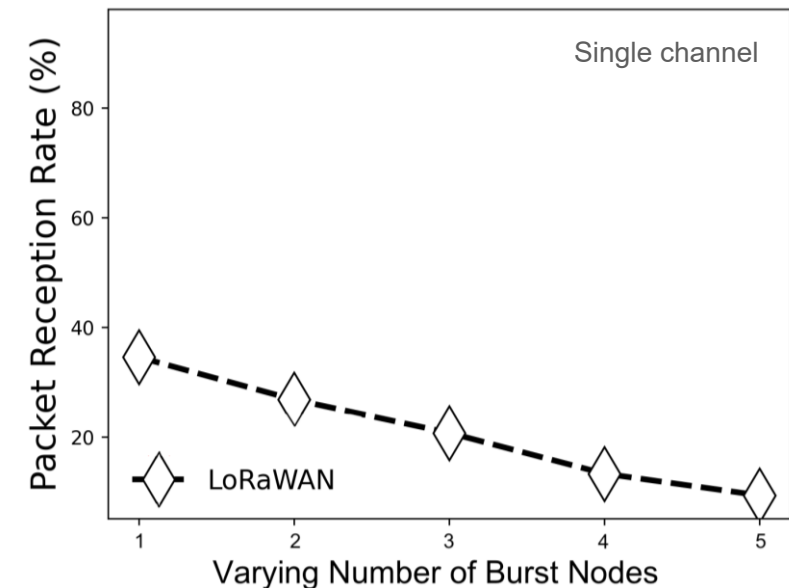
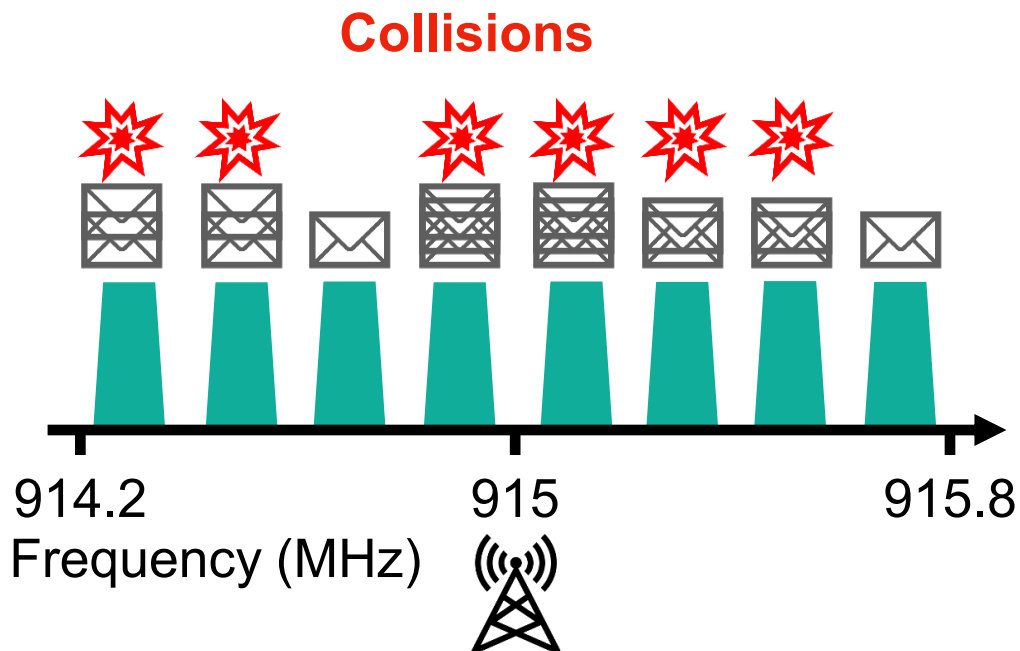


Fig: Packet Reception Rate for 5 nodes

*Validation from experiment in our lab

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Ensuring successful transmission of this traffic without loss is critical

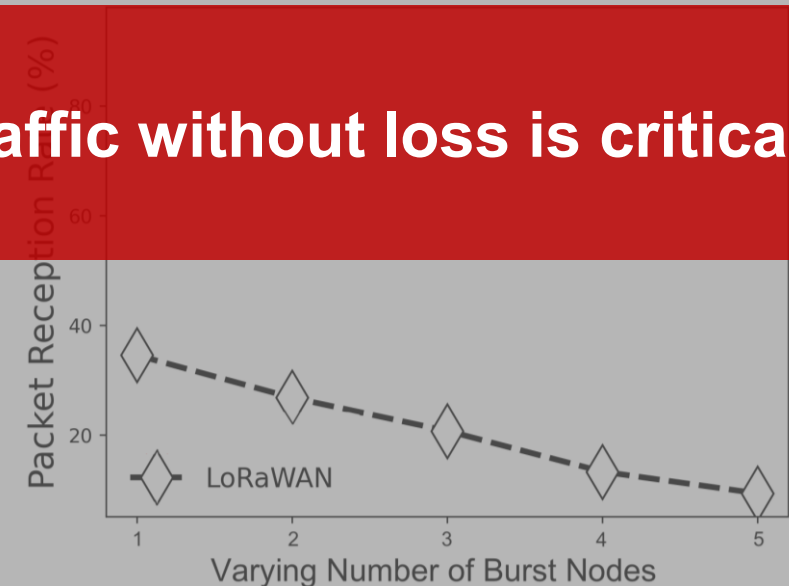
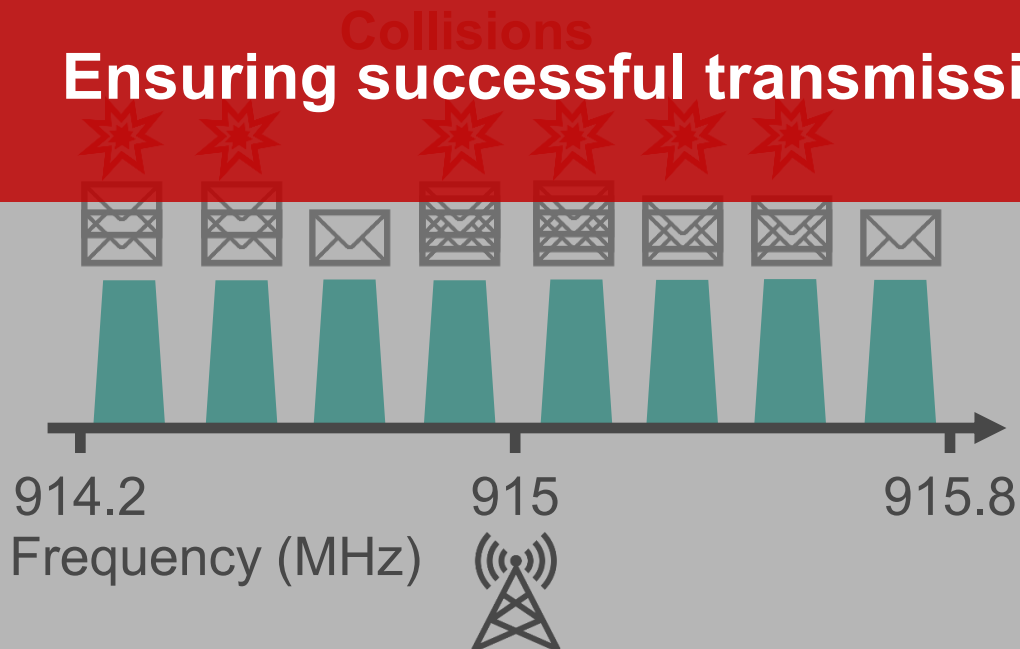


Fig: Packet Reception Rate for 5 nodes

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Burst-MAC

We introduce **Burst-MAC**, a low-overhead and bounded latency MAC protocol for LoRa networks to handle burst traffic.

◆ Semi-Distributed Scheduling Mechanism

- Nodes compute their Tx slots individually.
- Eliminates the need of centralized slot distribution.

◆ Devices switch to Burst-MAC only when burst is detected.

- Work in LoRaWAN mode otherwise.
- No time synchronization required during non-burst periods!

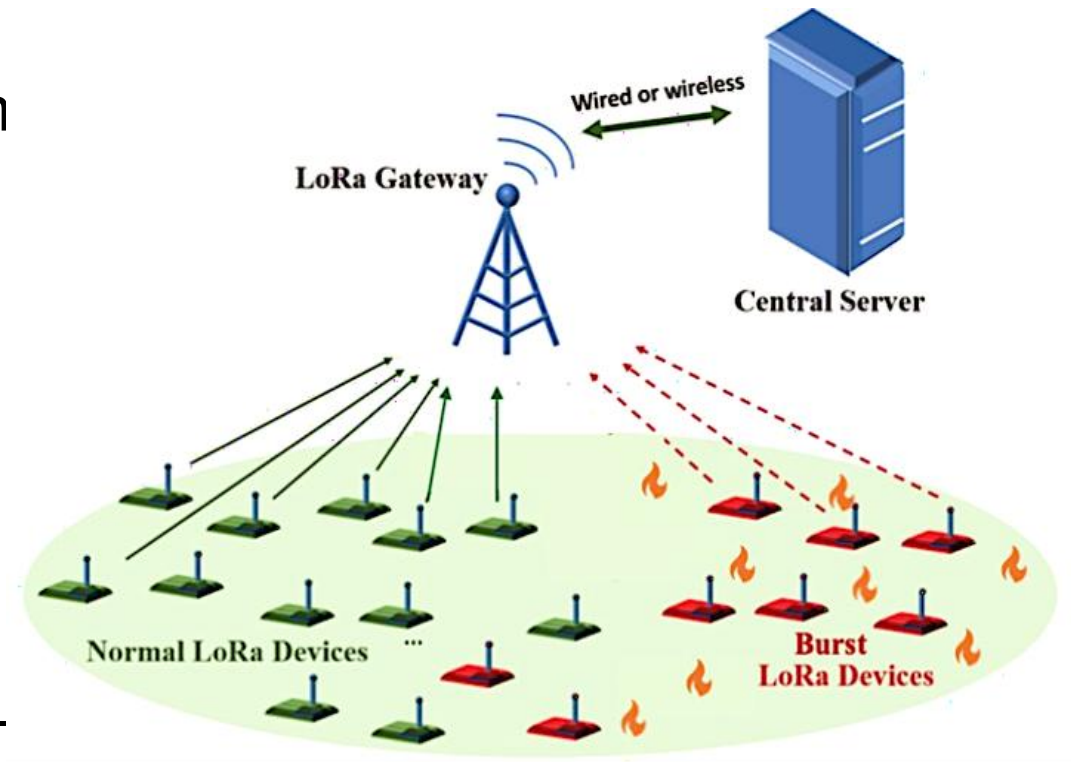
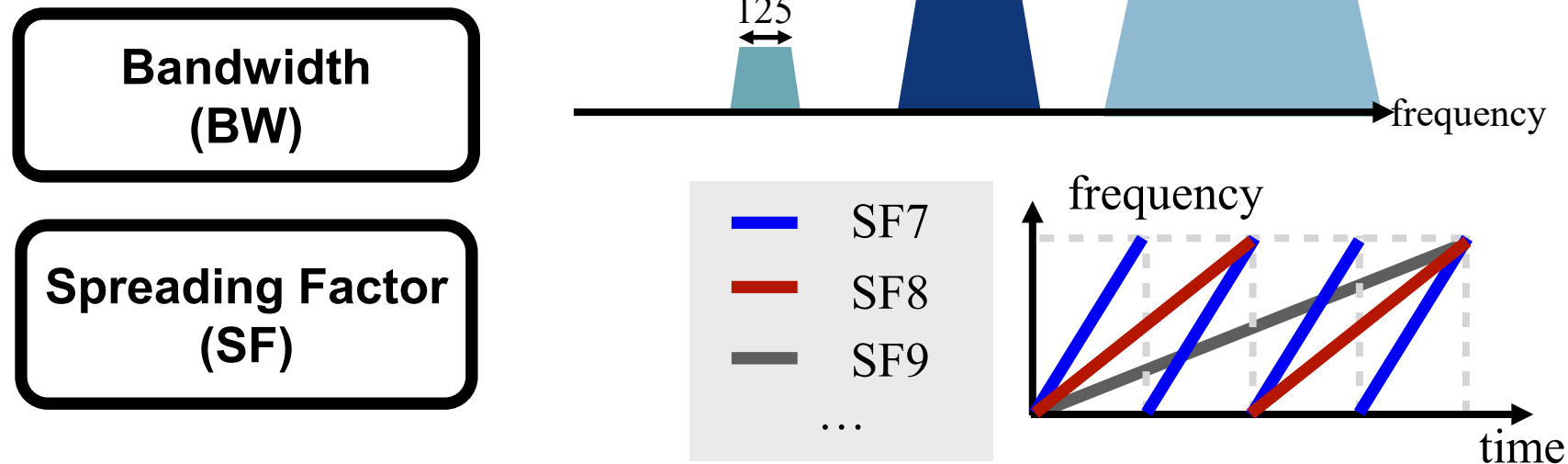


Fig: Nodes in the effected area undergo burst and switch to Burst-MAC. Nodes in non-effected areas work in LoRaWAN mode.

Burst-MAC

- ◆ Exploits **parallel packet reception** capability of LoRa gateway by using *Orthogonal Virtual Channels (VC)*.



<Physical Channel (Central Frequency, Bandwidth), Spreading Factor> → **Virtual channel**

e.g., <Ch 1, SF 7> -> VC 1

<Ch 1, SF 8> -> VC 2

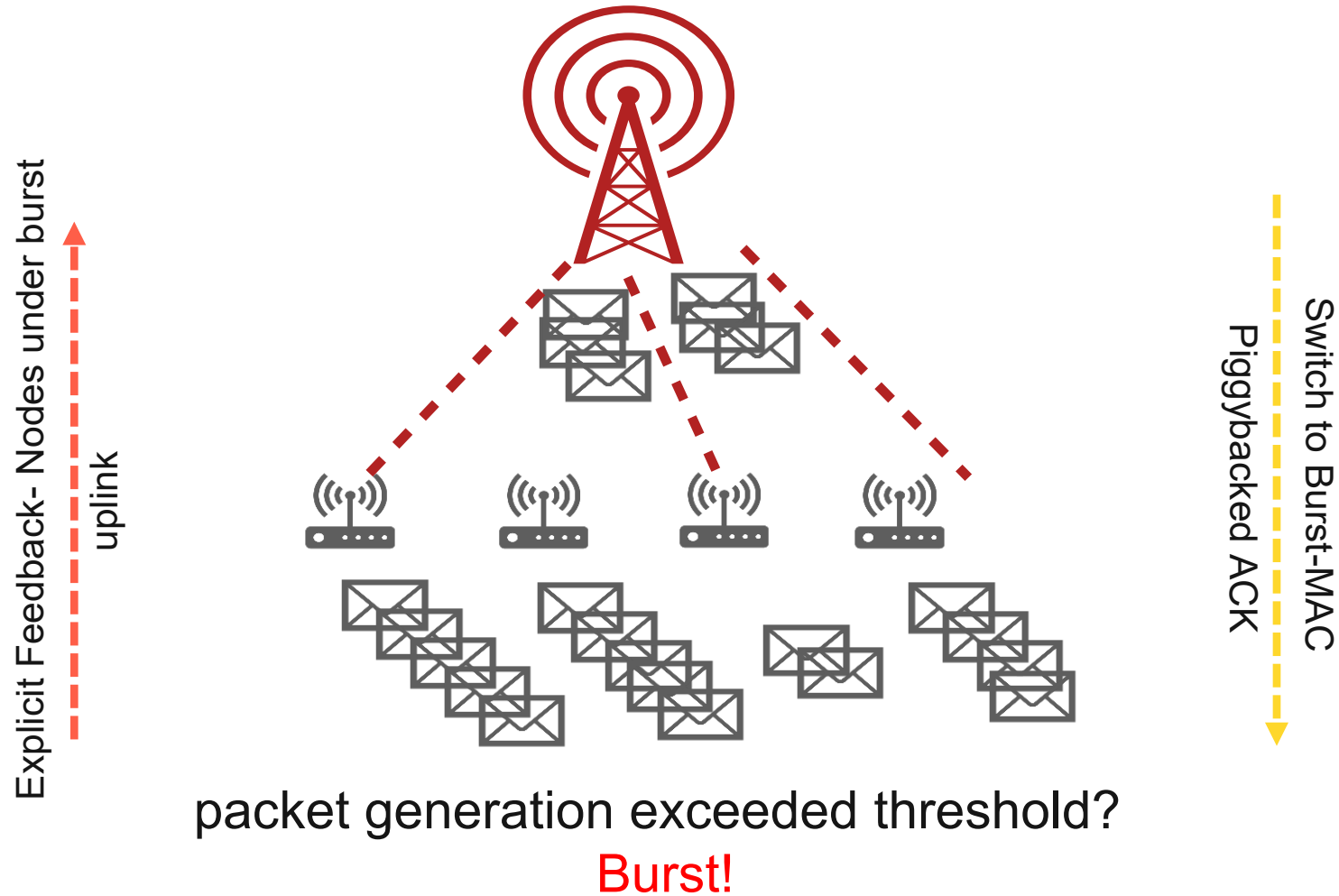
<Ch 4, SF 9> -> VC 3

Burst-MAC Design

Determining the Start of Burst Traffic

+ Collisions exceeded threshold?

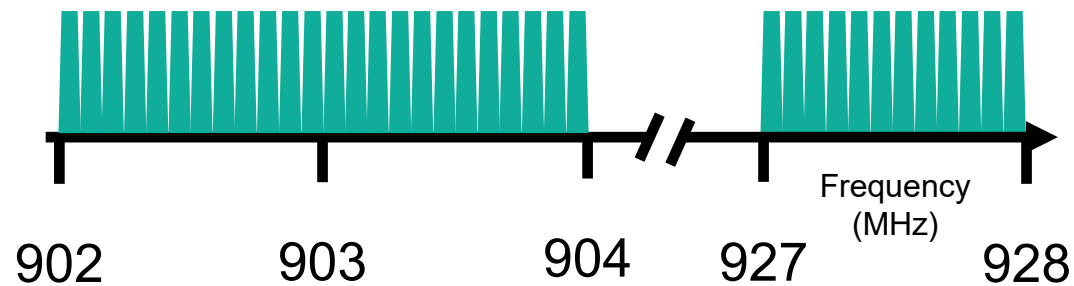
Burst Detected!



Burst-MAC Design

Grouping Nodes based on Virtual Channels

LoRa in the US, m_{up} uplink channels and s total number of SF



$$s = 6, m_{up} = 64$$

$$\text{Virtual Channels: } 6 \times 64 = 384$$

Hundreds of Virtual Uplink Channels.

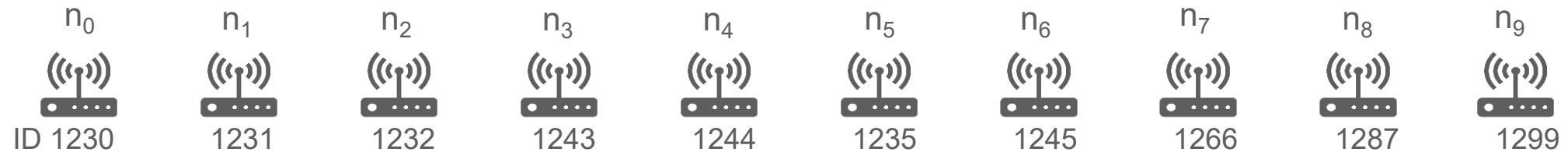
- ◆ Gateway groups the nodes based on their VC.
 - Only a few nodes per group.
- ◆ Each VC is a unique group.
 - Collisions confined within groups.
 - Enables concurrent transmissions.
 - Now collisions need to be resolved only inside each VC, which can be done using a transmission scheduling policy.

Burst-MAC Design

Hash-based Scheduling for Uplink Transmissions

Hash-based semi-distributed transmission scheduling policy!

- ◆ Nodes in burst mode within a VC individually compute their time slots for uplink transmission using a hash function, e.g.,



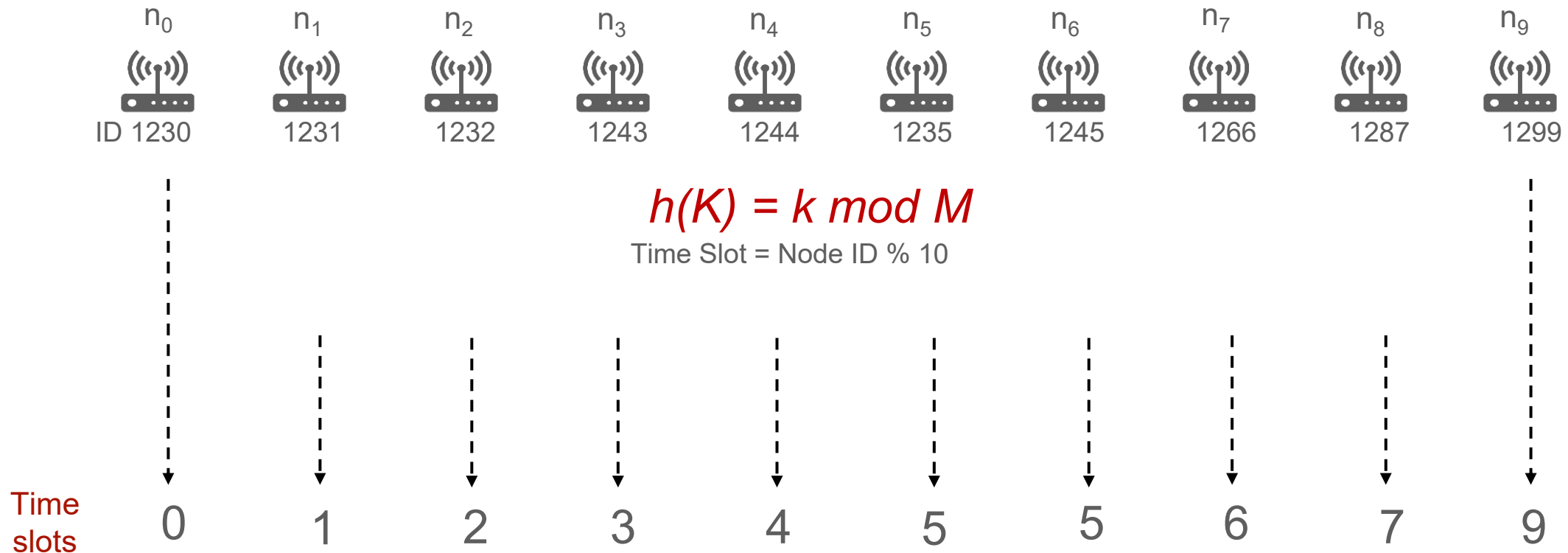
$$h(K) = k \bmod M$$

Hash value Node ID No. of nodes in a group

Hash value = node's uplink Tx time slot, *Superframe* defines the total no. of time slots for a group

Burst-MAC Design

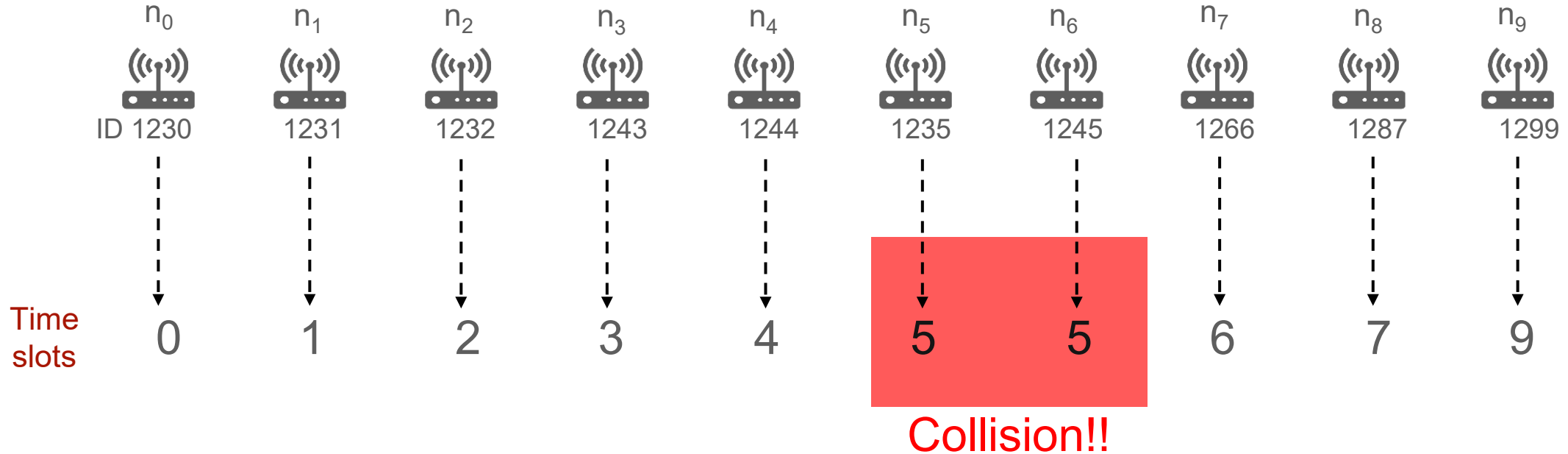
Hash-based Scheduling for Uplink Transmissions



Superframe length = 10

Burst-MAC Design

Ensuring Collision Free Transmissions

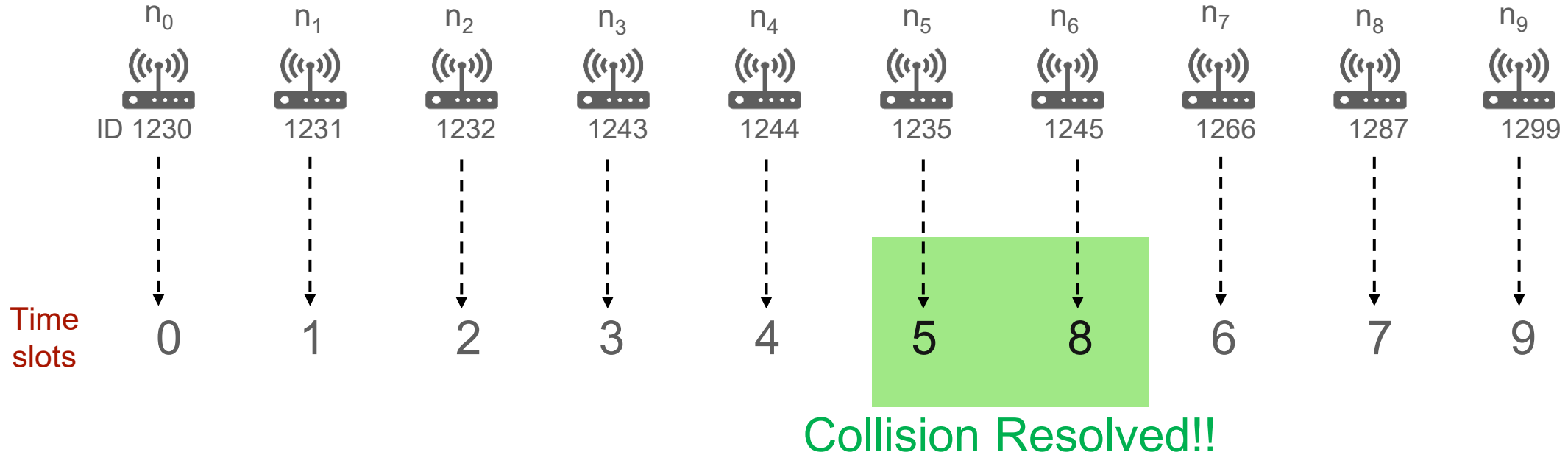


Burst-MAC **dynamically** handles time slot collisions.

New time slot via **Piggybacked ACK**

Burst-MAC Design

Ensuring Collision Free Transmissions



Burst-MAC maximizes overall utilization.

Ensures Bounded Latencies for every packet.

Before collision resolution: $2 * N_i$ time slots
After collision resolution: N_i time slots

Where,
 N_i = superframe length

Burst-MAC Design

Downlink Communication in Burst-MAC

- ✦ To ensure new schedules are received successfully, burst nodes work in **Class-B** mode.
- Nodes can receive ACKs in either RX1, RX2, or additional *ping slots*. More chances to receive downlink ACK containing new Tx slots.

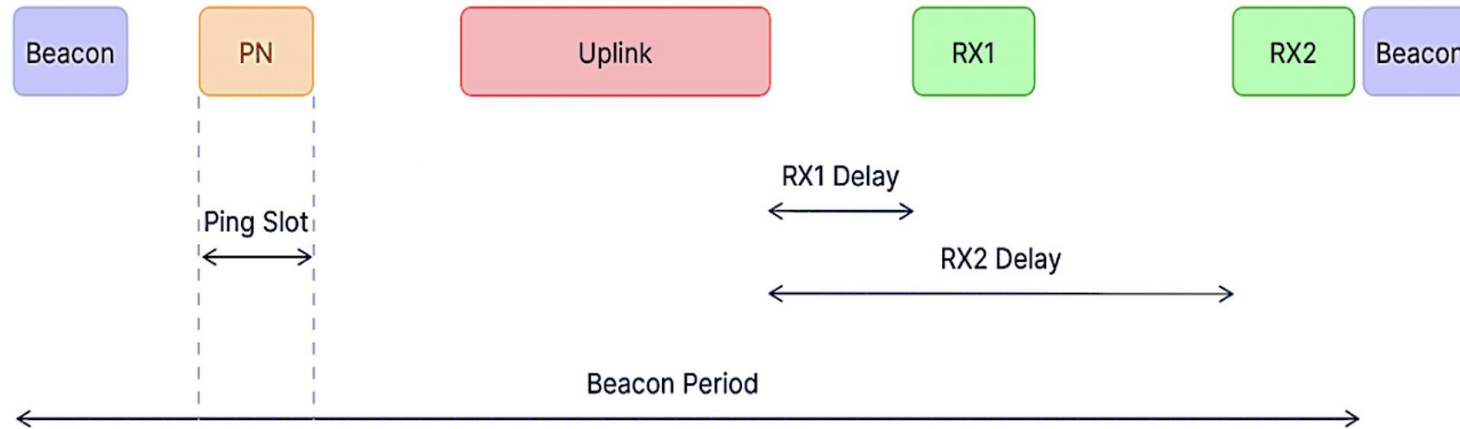
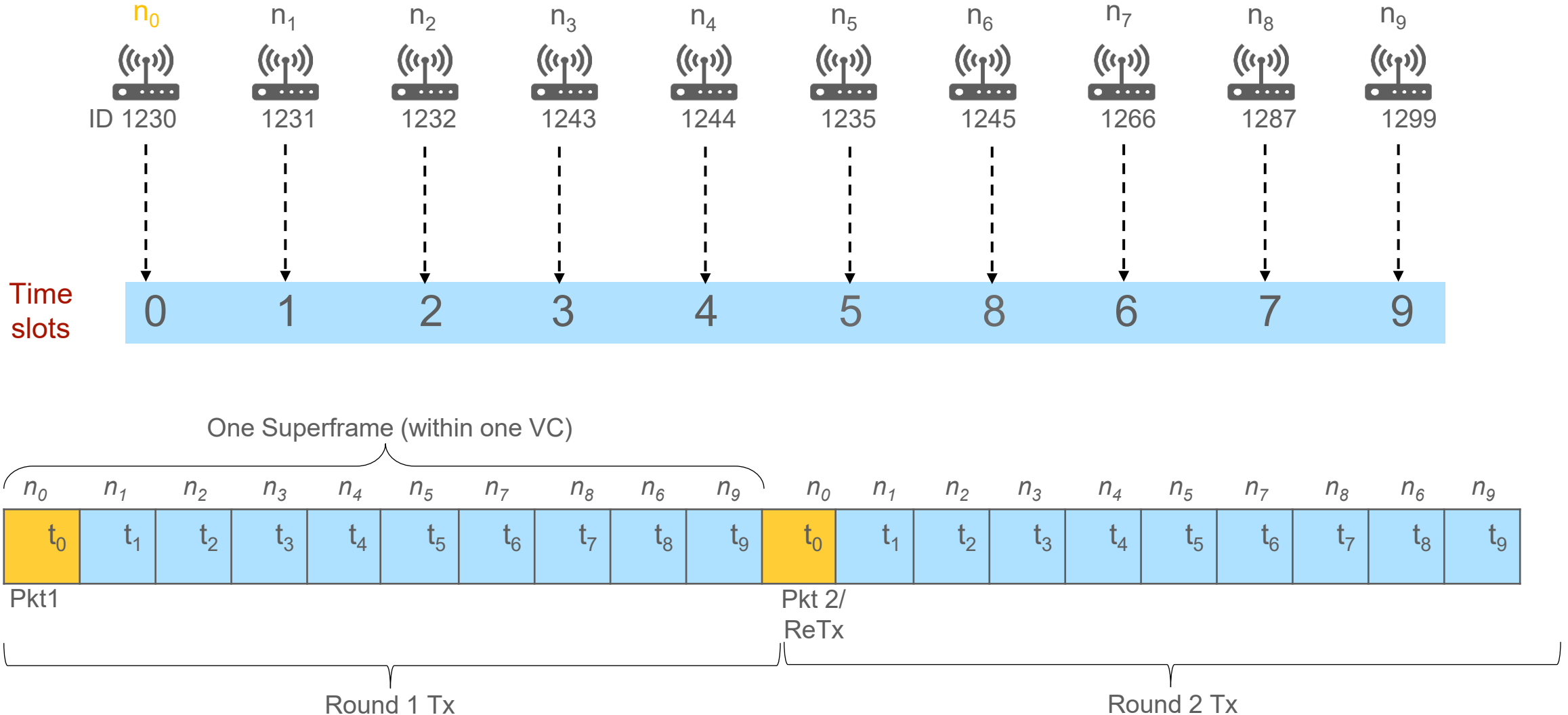


Fig: Class-B mode of LoRa

Increases chances of successful collision resolution

Burst-MAC Design

Final Collision Free Transmission Slots look like



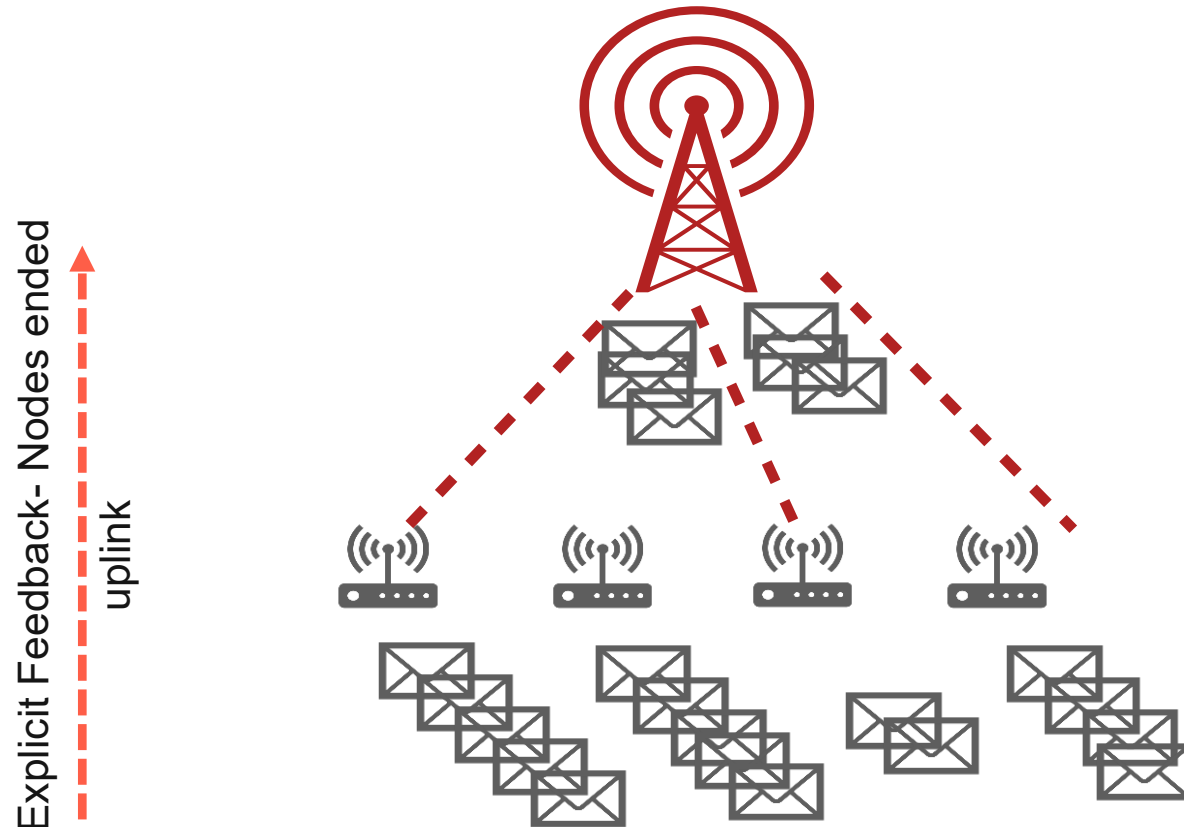
Burst-MAC Design

Handling the end of Burst Traffic

+ Increased no. of unused slots?

Burst End!

End Burst-MAC. Stop sending any schedules.



Transition from Class-B to Class-A. Stop computing tx slots.

Thank you!

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Read my paper at:

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